

Radon Mitigation Is Not the End of the Evidence

From Initial Measurement to Verified Environmental Consequence

INDEPENDENT TECHNICAL PRE-EXAMINATION | REFERENCE v2.0 | 10 OCTOBER 2026

STATUS: PRE-EXAMINATION | BASELINE NOT FROZEN | NO FORMAL DISPOSITION

Independent educational architecture demonstration. No endorsement, affiliation, approval, certification, or operational integration is claimed from the Indoor Environments Association (AARST), RadonAway, EPA, or any other institution. All illustrative cases, identifiers, readings, and decisions are hypothetical.

Governing question

Does this proposed consequence have sufficient Admissible Evidence, Applicable Authority, and Established Standing to become reality NOW?

Executive summary

A radon measurement is an observation, a mitigation installation is an intervention, and a post-mitigation test is a time- and condition-bounded verification. These events must remain distinguishable. TA-14 proposes a separate evidence-and-authority examination: preserving source records, establishing continuity, binding decisions to the right building and responsible parties, and limiting claims to what the evidence supports.

This document does not prescribe radon testing or mitigation, determine individual health risk, grant regulatory clearance, or replace qualified professionals or applicable requirements.

1. Examination scope and consequence

The candidate consequence must be stated precisely: approval to conduct work, acceptance that work was performed, a claim that mitigation achieved a specified result under specified conditions, or an ongoing-performance statement. These are different claims and require different evidence. A single result does not establish permanent conditions.

2. External standards and authority boundary

Project-specific testing methods, action levels, professional qualifications, required follow-up, and reporting obligations must be determined using current applicable law and standards. The following is an applicability checklist, not a claim of compliance or a standards crosswalk.

Source / authority	Project-specific verification
EPA radon guidance	Determine which current EPA consumer, testing, mitigation, and follow-up guidance applies to the setting.

Source / authority	Project-specific verification
ANSI/AARST standards	Identify current relevant measurement and mitigation standards and exact editions; confirm scope and adoption.
State / local authorities	Verify licensing, certification, permitting, disclosure, testing, and inspection obligations for the actual jurisdiction.
Owner / contract	Document property authority, work approval, acceptance rights, and ongoing responsibilities.

3. Eight-anchor TA-14 examination

Anchor	Question the record must answer
REALITY	What physical building and environmental condition actually existed?
RECORD	What attributable original measurement, context, and metadata exist?
CONTINUITY	Can custody, versions, transformations, and handoffs be traced?
ADMISSIBILITY	Is this evidence fit for this particular claim and decision?
BINDING	Is the evidence bound to the correct site, zone, scope, person, and authority?
COMMIT	What bounded disposition is recorded: ALLOW / HOLD / DENY / ESCALATE?
EXECUTION	Was the authorized intervention actually carried out within scope?
OUTCOME	What was verified, when, with what limits and revalidation triggers?

4. Hypothetical building case - evidence sequence

The following fictional example is intended to pressure-test the architecture, not illustrate a compliant radon measurement protocol. Numbers are deliberately omitted because method, duration, units, and applicable thresholds must be established for the actual jurisdiction and project.

Stage	Candidate record	Examination observation
Initial condition	RAD-OBS-001; source report and sampling metadata	Attributable result and location are available; interpretation remains bounded.
Proposed work	RAD-DEC-002; owner approval and contractor scope	Work authority must be established before execution.
Installation	RAD-WRK-003; as-built documentation	Supports work completion, not proof of environmental outcome.

Stage	Candidate record	Examination observation
Follow-up	RAD-VER-004; report referenced but missing zone identifier	Outcome cannot reliably be bound to treated zone.
Later change	RAD-CHG-005; ventilation configuration revised	Prior result may require re-examination for a new claim.

5. Illustrative HOLD decision

Claim examined: “The treated zone has a verified post-mitigation result.” Evidence gap: RAD-VER-004 lacks reliable zone attribution. Disposition: HOLD, for teaching purposes only. The record supports that mitigation work was documented, but not the stated outcome for the specific zone. Release condition: obtain appropriate qualified verification with defensible attribution and re-examine. This is not an actual TA-14 ruling.

6. Evidence register - minimum examination fields

Field	Illustrative contents
Identity	Site ID, zone, sampling point, relevant building configuration.
Measurement	Original report, dates, duration, method, device/laboratory, units, quality information.
Provenance	Creator, custody, receipt date, immutable source reference, version, transformation history.
Decision	Proposed consequence, evidence relied on, owner/approver, scope, constraints.
Work	Qualified party, authorization, as-built scope, completion record, deviations.
Verification	Post-work test and conditions, qualified interpretation, limitations, follow-up requirements.
Change	Equipment, pressure, ventilation, use, occupancy, and other material changes.

7. Responsibility and decision-rights matrix

Role	Boundary to establish
Testing professional / laboratory	Measurement method, data validity, attribution, reporting, and limitations.
Mitigation professional	Qualifications, proposed design, approved scope, installation and work records.
Owner / authorized representative	Permission, access, contract approval, acceptance, maintenance responsibilities.
Authority having jurisdiction	Applicable statutory and regulatory requirements, if any.
TA-14 examination layer	Evidence continuity and decision-boundary assessment; no statutory or professional authority presumed.

8. Disposition logic

Disposition	Illustrative decision condition
ALLOW	Specified claim has sufficient admissible evidence, applicable authority, and established standing within scope.
HOLD	Remediable gap prevents the specific claim from proceeding now.
DENY	Established conflict, disqualifying condition, or unsupported authority prevents the claim.
ESCALATE	Question requires another competent decision-maker, specialist, or authority.

A disposition attaches to a defined consequence, evidence baseline, authority context, and time. It is not a universal status of the building. No formal disposition is issued in this reference.

9. Changed-context revalidation

A satisfactory test at one moment does not prove perpetual performance. A material change in ventilation, pressure relationships, use, occupancy, building envelope, mitigation equipment, or test method can affect whether earlier evidence remains adequate for a later claim. Preserve original and subsequent measurements separately. Do not overwrite raw observations with revised models, corrections, or interpretations.

10. Candidate institutional demonstration

A proposed bounded pilot could use a synthetic or properly permissioned, de-identified case to examine the handoffs between testing, mitigation, verification, and subsequent change. A formal TA-14 examination would require a registered governance baseline, explicit scope, defined participants and decision rights, and a frozen candidate evidence package.

Pilot acceptance criteria

1. Original records and transformations are traceable. 2. The site and zone attribution is reproducible. 3. Decision rights are documented. 4. Missing verification produces an explicit HOLD rather than a fabricated pass. 5. A changed condition triggers a bounded re-examination. 6. The final package distinguishes observed facts, professional interpretation, and TA-14 disposition.

11. Examination worksheet

Prompt	Candidate response / evidence ID
Exact proposed consequence	_____
Site, zone, time and scope	_____
Original evidence and versions	_____
Known gaps and uncertainty	_____
Applicable authority / standing	_____
Proposed disposition and rationale	_____
Conditions and revalidation trigger	_____

12. Eight-image showroom alignment

Image	Teaching objective
01 - The Invisible Condition	Observation is not permanent characterization.
02 - The Original Record	Raw result and contextual metadata remain identifiable.
03 - Evidence Continuity	Provenance survives handoffs and transformations.
04 - The Mitigation Decision	Decision rights precede consequential work.
05 - Installation Is Not Verification	As-built work does not prove environmental effectiveness.
06 - The Post-Mitigation Test	Verification is attributable and bounded.
07 - Changed Conditions	Re-examination follows material changes.
08 - Authorized Environmental Consequence	Only the supported claim proceeds NOW.

Limitations and verification note

This reference is not a measurement protocol, mitigation design, certification, regulatory determination, exposure assessment, or statement that TA-14 has deployed a radon-specific operational integration. External requirements and editions must be verified before any real-world use. TA-14 registration does not confer regulatory or professional authority.

Institutional invitation: Qualified practitioners and organizations may consider a bounded, independently scoped registered-governance demonstration to test evidence continuity and decision boundaries without implying endorsement or altering existing professional responsibilities.